01 · LESSON

# In and out: surface area, diffusion and osmosis

*9C Threat from disease · Lesson 7 of 7*

Date _____ Class _____

**Today I can**

- ☐ I can calculate a surface area : volume ratio and explain why it matters for cells and organisms.
- ☐ I can describe how osmosis happens through a partially permeable membrane and use it to explain what I see.

**What happens today**

1. Do Now: a 1 cm cube
2. Learn: Surface area : volume ratio · Why the ratio matters · and more
3. Practical: Osmosis in potato strips
4. Challenge, then the exit ticket

Student Book pages 50 to 52 · Workbook pages 36 to 38

**What I learned today (one or two sentences)**



02 · KEY VOCABULARY

# Key Vocabulary

| Term | Definition |
|---|---|
| **surface area** | the total area of the outside of an object |
| **volume** | the amount of space an object takes up |
| **ratio** | one quantity compared with another, e.g. 6 : 1 |
| **diffusion** | the overall movement of particles from where they are more concentrated to where they are less concentrated |
| **concentration gradient** | a difference in concentration between two places |
| **osmosis** | the diffusion of water molecules through a partially permeable membrane |
| **partially permeable membrane** | a membrane with tiny holes that let small molecules through but not large ones |
| **cell membrane** | the partially permeable layer around a cell |
| **unicellular** | made of one cell |
| **villi** | finger-like folds that give the small intestine a large surface area |
| **pandemic** | an outbreak of a disease across the whole world |



03 · QUICK REVIEW

# Quick Review

*a 1 cm cube*

A cube has sides of 1 cm.

**1** Calculate its surface area. (It has 6 faces.)

**2** Calculate its volume.

**3** Which gas diffuses into cells so they can respire?



04 · CORE

# Core 1 of 2

*Everyone does these. Answer on the card or under it.*

**1** Complete the text using words from the word bank.

The movement of all particles is _____. Where there is a concentration _____, more particles move out from a region of _____ concentration than from a region of _____ concentration. This spreading out is known as _____.

**2** Complete the table for three cubes.

| Cube | Surface area (cm2) | Volume (cm3) | Surface area : volume ratio |
|---|---|---|---|
| sides 1 cm | 6 | 1 | 6 : 1 |
| sides 2 cm | | 8 | |
| sides 3 cm | 54 | | |

# Core 2 of 2

*Everyone does these. Answer on the card or under it.*

**3** Potato strips in water and salt solution.

**a)** Record your results.

| Strip | Liquid | Length at start (mm) | Length at end (mm) | Change (mm) | Firm or floppy? |
|---|---|---|---|---|---|
| A | pure water | | | | |
| B | salt solution | | | | |

**b)** Tick the strip that gained water.

- ☐ A strip A (water)
- ☐ B strip B (salt solution)

**c)** Explain your results using the word osmosis.





# Challenge 1 of 3

*When Core is finished. Full sentences.*

**C1** A silky anteater has a mass of about 400 g and eats up to 5000 ants a day. A giant anteater has a mass of about 40 kg and eats up to 30 000 ants a day.

**a)** Calculate the number of ants eaten per gram of body mass for each anteater.

**b)** Suggest a reason for the difference, using ideas about surface area : volume ratio.





# Challenge 2 of 3

*When Core is finished. Full sentences.*

**C2** Osmosis in living things.

**a)** Paramecium is a unicellular organism that lives in fresh water. It has a contractile vacuole that collects water from the cytoplasm and pumps it out. Explain why Paramecium needs to remove water from its cytoplasm.





# Challenge 3 of 3

*When Core is finished. Full sentences.*

**C2** *continued*

**b)** Tropical sprue stops food being digested properly, so undigested food stays dissolved inside the small intestine. Explain how this can cause diarrhoea (watery faeces) by osmosis.

# Exit Ticket

*Hand this card in at the end of the lesson.*

Name _____

**1** What is osmosis?

**2** A 2 cm cube has a surface area of 24 cm2 and a volume of 8 cm3. State its surface area : volume ratio.

**3** Give one reason why rapid international travel makes a pandemic more likely.

**Today I can:** Goal 1 ☐ Goal 2 ☐